MOTIVIC HOMOTOPY THEORY

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1. Introduction

Throughout this section, let k be a field and let SmProp_k be the category of smooth proper k-schemes.

Grothendieck's motives. Grothendieck's proposed the concept of *motives* in the 1960s to explain the common features of the *Weil cohomology* theories of algebraic varieties.

Example 1.1 (Weil cohomology theories). Let $X \in \mathrm{SmProp}_k$.

(a) (Betti cohomology) Let $\iota\colon k \hookrightarrow \mathbf{C}$ be an embedding. Consider the *singular cohomology* of $X(\mathbf{C})$ with the analytic topology $\mathrm{H}^*_{\mathrm{Betti}}(X) = \mathrm{H}^*(X(\mathbf{C}), \mathbf{Q})$. This defines a functor
$$\mathrm{H}^*_{\mathrm{Betti}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_{\mathbf{Q}}^{\mathrm{gr}}$$
depending on the embedding.

(b) (Algebraic de Rham cohomology) Suppose that $\mathrm{char}(k) = 0$. The *algebraic de Rham complex* defines cohomology groups $\mathrm{H}^*_{\mathrm{dR}}(X/k)$ equipped with the Hodge filtration. There are functors
$$\mathrm{H}^*_{\mathrm{dR}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_k^{\mathrm{gr}},$$
where $\mathrm{FilVect}_k$ is the category of finite-dimensional filtered k-vector spaces.

(c) (Étale cohomology) Let k^{sep} be a separable closure of k, $G_k = \mathrm{Gal}(k^{\mathrm{sep}}/k)$, and let $\ell \neq \mathrm{char}(k)$ be a prime. The *étale cohomology* groups $\mathrm{H}^*_{\text{ét}}(X_{k^s}, \mathbf{Q}_\ell)$ give us the functors
$$\mathrm{H}^*_{\text{ét}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}_{\mathbf{Q}_\ell}^{\mathrm{gr}}.$$
This construction does not require an embedding into $\mathbf{C}$ and applies in positive characteristic.

(d) (Crystalline cohomology) Let k be a perfect field of characteristic $p > 0$. The *crystalline cohomology* groups $\mathrm{H}^*_{\mathrm{cris}}(X/\mathrm{W}(k))[\frac{1}{p}]$ give us the functors
$$\mathrm{H}^*_{\mathrm{cris}}\colon \mathrm{SmProp}_k^{\mathrm{op}} \to \mathrm{Vect}^{\mathrm{gr}}_{\mathrm{W}(k)[\frac{1}{p}]},$$

Certain features suggest that these cohomology theories can be compared, at least to some extent.

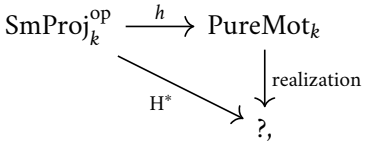
Remark 1.2 (Comparison between Weil cohomology theories). The Weil cohomology theories in Example 1.1 are related by comparison isomorphisms. Let $k \hookrightarrow \mathbf{C}$ and $k^{\mathrm{sep}} \hookrightarrow \mathbf{C}$ be embeddings. There are natural isomorphisms
$$\mathrm{H}^*_{\mathrm{Betti}}(X) \otimes_{\mathbf{Q}} \mathbf{Q}_\ell \simeq \mathrm{H}^*_{\text{ét}}(X_{k^{\mathrm{sep}}}, \mathbf{Q}_\ell) \quad \text{and} \quad \mathrm{H}^*_{\mathrm{Betti}}(X) \otimes_{\mathbf{Q}} \mathbf{C} \simeq \mathrm{H}^*_{\mathrm{dR}}(X/k) \otimes_k \mathbf{C}.$$
The difficulty lies in the fact that the theories do not take values in vector spaces over a common field.

Further evidence is provided by numerical invariants. For example,

Theorem 1.3 (Independence of ℓ)**.** *Let X be a smooth and proper scheme over k. The i-th* Betti number
$$b_i(X) = \dim_{\mathbf{Q}_\ell} \mathrm{H}^i_{\text{ét}}(X_{k^s}, \mathbf{Q}_\ell)$$
is independent of the prime $\ell \neq \mathrm{char}(k)$.

Remark 1.4. It remains an open question whether Theorem 1.3 extends to arbitrary schemes of finite type over a field. This is one of the celebrated *independence-of-ℓ conjectures*.

Grothendieck's idea was to account for these similarities by a category of *pure motives* over $\mathbf{Q}$, together with realization functors recovering the different cohomology theories. The proposed picture is



where ? is understood as the cohomological coefficients and $h(X)$ is a universal linear object attached to X.

Tannakian duality asserts that every symmetric monoidal category satisfying the Tannakian axioms is equivalent to the category of representations of an affine *gerbe*. The category of pure motives is expected to be *Tannakian* over $\mathbf{Q}$ and satisfies $\mathrm{PureMot}_k \simeq \mathrm{Rep}(\mathrm{Gal}(k)^{\mathrm{mot}})$, where the right hand side denotes the representation of gerbs of motivic Galois group.the *motivic Galois group*. The cohomological invariants discussed above are understood as different realizations of underlying objects in the category of motives.

Although there are some possible categories for pure motives, their expected Tannakian properties remain conjectural in general. Grothendieck's *standard conjectures* on algebraic cycles would supply the necessary compatibility with cohomological realizations.

Although the theory of pure motives was originally developed in part to address the Weil conjectures, it remains a fruitful idea in algebraic geometry, even after the conjectures were proved by other methods.

Stable homotopy theory. We now pass from graded cohomology groups to complexes and spectra. This suggests a homotopical refinement of the motivic viewpoint. A "cohomology theory" on a "geometric category" $\mathcal{G}$ is a contravariant functor

$$E\colon \mathcal{G}^{\mathrm{op}} \longrightarrow \mathcal{C},$$

where $\mathcal{C}$ is a stable category. Examples of $\mathcal{G}$ include topological spaces, manifolds, and schemes; examples of $\mathcal{C}$ include spectra Sp, module spectra, filtered spectra, and the derived categories $\mathrm{D}(\mathbf{Z})$ and $\mathrm{D}(k)$.

The following are some examples of cohomology theories:

Example 1.5 (Cohomology theories)*.*

(a) (Ordinary cohomology) Singular cochains define a functor

$$\mathrm{C}^*(-,\mathbf{Z})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{D}(\mathbf{Z}).$$

For a locally contractible paracompact Hausdorff space X, this agrees with $\mathrm{R}\Gamma(X,\mathbf{Z}_X)$, the derived global sections of the constant sheaf.

(b) (Complex topological K-theory) The spectrum KU defines a functor

$$\Gamma(-,\mathrm{KU})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{Sp}, \quad \Gamma(X,\mathrm{KU}) = \mathrm{map}_{\mathrm{Sp}}(\Sigma^\infty_+ X, \mathrm{KU}),$$

its homotopy group gives us the *complex topological K-theory* groups

(c) (Complex cobordism) The *complex cobordism spectrum* MU defines

$$\Gamma(-,\mathrm{MU})\colon \mathrm{Top}^{\mathrm{op}} \to \mathrm{Sp}, \quad \Gamma(X,\mathrm{MU}) = \mathrm{map}_{\mathrm{Sp}}(\Sigma^\infty_+ X, \mathrm{MU}).$$

its homotopy groups gives us the complex cobordism groups.

(d) (Étale cohomology) Let ℓ be a prime. On schemes on which ℓ is invertible, we have

$$\mathrm{R}\Gamma_{\text{ét}}(-,\mathbf{Z}/\ell)\colon \mathrm{Sch}^{\mathrm{op}}_{\mathbf{Z}[\frac{1}{\ell}]} \to \mathrm{D}(\mathbf{Z}/\ell).$$

Let X be a scheme of finite type over k, where $\mathrm{char}(k) \neq \ell$. Its ℓ-adic cohomology complex are

$$\mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}_\ell) = \lim_{n\geq 1} \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}/\ell^n), \quad \text{and} \quad \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Q}_\ell) = \mathrm{R}\Gamma_{\text{ét}}(X,\mathbf{Z}_\ell) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell,$$

(e) (Algebraic K-theory) Let $\mathrm{Sch}_{\mathrm{qcqs}}$ be the category of quasi-compact quasi-separated schemes. *Nonconnective algebraic K-theory* gives a functor

$$\mathrm{K}\colon \mathrm{Sch}^{\mathrm{op}}_{\mathrm{qcqs}} \to \mathrm{Sp}, \quad X \mapsto \mathrm{K}(\mathrm{Perf}(X)),$$

Let R be a commutative ring and Vect_R be the category of finitely generated projective R-modules. Then

$$\Omega^\infty \mathrm{K}(\mathrm{Spec}(R)) \simeq (\mathrm{Vect}_R^{\simeq})^{\mathrm{grp}},$$

where $(-)^{\mathrm{grp}}$ is the group completion functor.

In fact, cohomology theories in the examples above carries more structure than just functoriality.

Example 1.6 (Additional structures). Let X be a topological space.

(a) (Cup product) The cup product is a map
$$- \smile -\colon \mathrm{H}^i(X,\mathbf{Z}) \otimes \mathrm{H}_j(X,\mathbf{Z}) \to \mathrm{H}_{j-i}(X,\mathbf{Z}),$$
which makes $\mathrm{H}^*(X,\mathbf{Z})$ into a graded-commutative ring.

(b) (Twisted coefficients) Let A be a local system of abelian groups on X. There is the notion of cohomology $\mathrm{H}^*(X,A)$ and homology $\mathrm{H}_*(X,A)$ with coefficients in A.

(c) (Wrong-way functoriality) Let $f\colon Y \to X$ be a proper smooth submersion of manifolds without boundary, of relative dimension d, and let or_f denote its relative orientation local system. Integration along the fibers defines a *Gysin pushforward*
$$f_!\colon \mathrm{H}^{i+d}(Y,\mathrm{or}_f) \longrightarrow \mathrm{H}^i(X,\mathbf{Z}).$$
A choice of relative orientation identifies $\mathrm{or}_f \simeq \mathbf{Z}_Y$, yielding an untwisted pushforward.

Example 1.7 (Poincaré duality). Let X be an n-manifold without boundary, with orientation local system or_X. Poincaré duality gives
$$\mathrm{H}^i(X,\mathbf{Z}) \simeq \mathrm{H}^{\mathrm{BM}}_{n-i}(X,\mathrm{or}_X), \qquad \mathrm{H}^i_c(X,\mathbf{Z}) \simeq \mathrm{H}_{n-i}(X,\mathrm{or}_X),$$
where $\mathrm{H}^{\mathrm{BM}}_*$ denotes *Borel–Moore homology*. The dualizing complex is $\omega_X \simeq \mathrm{or}_X[n]$.

Example 1.8 (G-theory). Let X be a regular noetherian scheme of finite Krull dimension. The canonical map
$$\mathrm{K}(X) \simeq \mathrm{G}(X).$$
is an isomorphism. This is a K-theoretical analogue of Poincaré duality.

Six functor formalism. Grothendieck's fundamental insight was to organize cohomology theories categorically, replacing cohomology groups by categories of coefficients that are *bivariantly functorial*.

The category of coefficients for ordinary cohomology on a topological space X is the category of sheaves $\mathrm{Shv}(X,\mathrm{D}(\mathbf{Z}))$. For every continuous map $f\colon Y \to X$, there is a *pullback* functor
$$f^*\colon \mathrm{Shv}(X,\mathrm{D}(\mathbf{Z})) \to \mathrm{Shv}(X,\mathrm{D}(\mathbf{Z})), \quad (f_*\mathcal{F})(U) = \mathcal{F}(f^{-1}(U)).$$
where $U \subseteq X$ is an open subset. It admits a right adjoint f_*, i.e., the *pushforward* functor.

Suppose that Y and X are locally compact Hausdorff spaces. There is the *pushforward of proper support* $f_!\colon \mathrm{Shv}(Y,\mathrm{D}(\mathbf{Z})) \to \mathrm{Shv}(X,\mathrm{D}(\mathbf{Z}))$. The formula for $f_!$ is slightly more complicated:
$$(f_!\mathcal{F})(U) = \operatorname*{colim}_{K \subseteq f^{-1}(U)} \mathrm{fib}\left(\mathcal{F}\left(f^{-1}(U)\right) \to \mathcal{F}\left(f^{-1}(U) \setminus K\right)\right),$$
where the colimit is taken over all proper subsets over U. The functor $f_!$ admits a right adjoint $f^!$, the *exceptional pullback functor*

The above datum are related by the *base change isomorphism*.

Theorem 1.9 (Proper base change)**.** *Consider a cartesian square of locally compact Hausdorff spaces*
$$\begin{array}{ccc} Y' & \xrightarrow{h} & Y \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ X' & \xrightarrow{g} & X. \end{array}$$
There exists a canonical isomorphism $g^* f_! \simeq f'_! h^*$.

Remark 1.10. Let X be a locally compact Hausdorff space, and let $p\colon X \to *$.

(a) The *cohomology* of X with coefficients in $\mathbf{Z}$ is given by the pushforward $\Gamma(X,\mathbf{Z}) = p_*\mathbf{Z}$.

(b) The *cohomology with compact support* of X with coefficients in $\mathbf{Z}$ is given by the pushforward with proper support $\Gamma_c(X,\mathbf{Z}) = p_!\mathbf{Z}$.

(c) Writing $\omega_X = p^!\mathbf{Z}$ for the dualizing sheaf, the *homology* of X with coefficients in $\mathbf{Z}$ is given by $p_!\omega_X = p_!p^!\mathbf{Z}$.

The above datum can be incorperated into the category of *spans*.

Definition 1.11 (Category of spans)**.** Let $\mathcal{G}$ be a category with finite limits. Its *category of spans* $\mathrm{Span}(\mathcal{G})$ is defined as follows:

(a) the objects are those of $\mathcal{G}$;
(b) a morphism $X \to Y$ is an isomorphism class of spans $X \xleftarrow{f} Z \xrightarrow{g} Y$.
(c) the composite of $X \leftarrow Z \to Y$ and $Y \leftarrow T \to W$ is represented by $X \leftarrow Z \times_Y T \to W$.
(d) the identity of X is represented by $X \xleftarrow{\mathrm{id}} X \xrightarrow{\mathrm{id}} X$.

Let LCH be the category of locally compact Hausdorff spaces. The sheaf operations give a functor

$$\mathrm{Shv}(-, \mathrm{D}(\mathbf{Z})) \colon \mathrm{Span}(\mathrm{LCH}) \longrightarrow \widehat{\mathrm{Cat}}, \quad \left(X \xleftarrow{f} Z \xrightarrow{g} Y\right) \mapsto g_!f^*.$$

Its compatibility with composition is exactly the compatibility expressed by base change.

Definition 1.12 (Presentable six-functor formalism)**.** Let $\mathcal{G}$ be a category with finite limits. A *presentable six-functor formalism* on $\mathcal{G}$ is a lax symmetric monoidal functor

$$\mathrm{Span}(\mathcal{G}) \longrightarrow \mathrm{Pr}^L,$$

where Pr^L is the *category of presentable categories* and colimit-preserving functors, with the *Lurie tensor product*.

The topological example gives a presentable six-functor formalism, i.e., there exists a functor

$$\mathrm{Shv}(-, \mathrm{Sp}) \colon \mathrm{Span}(\mathrm{LCH}) \longrightarrow \mathrm{Pr}^L.$$

In algebraic geometry, two "universal" sources of six-functor formalisms are *motivic homotopy theory*, developed by Morel and Voevodsky, and the theory of *analytic stacks*, developed by Clausen and Scholze. The former encompasses many of the cohomology theories discussed above, including étale cohomology and variants of K-theory, whereas the latter provides a six-functor formalism for quasi-coherent sheaves.[1]

Motivic homotopy theory. Morel and Voevodsky's construction gives a functor

$$\mathrm{SH}(-) \colon \mathrm{Sch}^{\mathrm{op}} \longrightarrow \mathrm{Pr}^L, \qquad X \longmapsto \mathrm{SH}(X),$$

where $\mathrm{SH}(X)$ is the *stable* $\mathbf{A}^1$*-homotopy category* over X. Its objects are called *motivic spectra*. This is the algebraic analogue of the six functor formalism

$$\mathrm{Shv}(-, \mathrm{Sp}) \colon \mathrm{Top}^{\mathrm{op}} \longrightarrow \mathrm{Pr}^L, \quad X \longmapsto \mathrm{Shv}(X, \mathrm{Sp}).$$

Theorem 1.13 (Ayoub)**.** *On qcqs schemes,* SH *admits a six-functor formalism, i.e. a lax symmetric monoidal functor*

$$\mathrm{Span}_{\mathrm{sep,ft}}(\mathrm{Sch}_{\mathrm{qcqs}}) \longrightarrow \mathrm{Pr}^L, \qquad X \longmapsto \mathrm{SH}(X).$$

where the source is the category of spans whose right leg is separated and of finite type.

One goal of this course is to understand this formalism. Morel and Voevodsky's construction rests on three theorems: *homotopy purity* and *localization*, due to Morel and Voevodsky, and *ambidexterity*, due to Ayoub and Röndigs. The first two are *unstable* results.

In topology, one studies the relative manifolds over a topological space. The scheme-theoretical analogue of relative manifolds are *smooth schemes*.

Definition 1.14 (Relative manifolds)**.** Let X be a topological space. A *relative manifolds* over X is a map $Y \to X$ of topological spaces locally isomorphic to a projection $D^n \times U \to U$, where $U \subset X$ is an open subset.

[1] A natural question is whether these two approaches can be unified. Moreover, a related question is to what extent algebraic K-theory can be incorporated into the six-functor formalism of motivic homotopy theory. At present, the available formalism for algebraic K-theory is only partial, although a variant of the theory fits into this framework.

Stable homotopy theory provides a setting for cohomology theories. Unstable homotopy theory also retains nonabelian information, such as the fundamental group. Historically, this viewpoint was motivated in part by cohomology with nonabelian coefficients. In Definition 1.12, the category Pr^L accommodates such nonabelian information.

Passing to homotopy-invariant sheaves of animae on relative manifolds over X gives $\mathrm{Shv}(X, \mathrm{An})$. Analogously, for schemes, there is the category of $\mathbf{A}^1$*-invariant Nisnevich sheaves* of anima on smooth X-schemes, which is the nonstable analogue of $\mathrm{SH}(X)$.

Passing to homotopy-invariant sheaves on relative manifolds over X recovers $\mathrm{Shv}(X, \mathrm{An})$. Analogously, in algebraic geometry, the category $\mathrm{H}(X)$ of $\mathbf{A}^1$*-invariant Nisnevich sheaves* on smooth X-schemes provides the unstable counterpart of $\mathrm{SH}(X)$. To pass from $\mathrm{H}(X)$ to $\mathrm{SH}(X)$, one needs to add the Tate twist. Therefore, $\mathrm{SH}(X)$ is not the naive categorical stabilization of $\mathrm{H}(X)$.

The analogy can be summarized as follows:

Topological spaces	Schemes
$\mathrm{Shv}(X, \mathrm{Sp})$	$\mathrm{SH}(X)$
$\mathrm{Shv}(X, \mathrm{An})$	$\mathrm{H}(X)$
Relative manifolds over X	Smooth schemes over X
Ordinary cohomology H$\mathbf{Z}$	Motivic cohomology H$\mathbf{Z}$
Complex K-theory KU	Homotopy K-theory KGL
Complex cobordism MU	Algebraic cobordism MGL

Synthetic category theory. In these notes, we will freely speak of (higher) category theory. Here we will explain the basic ideas of homotopy theory and its relation to the foundations of mathematics.

Set theory provides a logical foundation, but its language does not directly reflect several features of mathematical practice.

First, every object of set theory is a set, and equality can be asked of any two objects. In classical logic, they are either equal or unequal. Mathematical language, by contrast, is usually *strongly typed*: one speaks of an integer, a group, or a manifold. *Type theory* makes this explicit by considering each term with a specified type. Unlike set theory, where any two elements can be collected into a set, type theory does not automatically place terms of different types in a common type.

Type conversions are nevertheless ubiquitous. A map between types sends a term of the source to a term of the target. For example, the integers and the even integers both appear to contain 2. In type theory, the inclusion of even integers into integers identifies the images of these terms in a common type. In this sense, the two types have an element in common.

Second, the usual formulation of set theory uses first-order logic, whose variables range over all sets. Mathematical statements generally use restricted quantifiers, such as "for every manifold X." Such restrictions can be expressed in set theory, whereas type theory incorporates them into the language.

Third, many collections of mathematical objects are not naturally set-like. If A is a set and $x, y \in A$, the equality $x = y$ is a truth value: classically, it is either true or false. If x and y are objects of a category, however, there may be several isomorphisms between them. Each gives a way of identifying the objects and transporting their categorical properties. If these identifications are regarded as equalities, equality carries more information than a truth value.

The homotopical viewpoint is to regard identifications themselves as mathematical objects. For x, y in such a collection A, there is a collection $\mathrm{Eq}(x, y)$ of identifications, which in turn has its own identifications. This is a fundamental principle of homotopy theory. An object with this structure is called an *anima*, a *space*, or an ∞*-groupoid*; in type theory, it is interpreted as a *type*. The collection of animae form a category An.

A set is a special case: for each $x, y \in A$, the anima $\mathrm{Eq}(x, y)$ is empty or contractible, expressing that equality is false or true. Here contractibility means uniqueness up to all higher identifications. An anima A is a *truth*

value if $\mathrm{Eq}(x, y)$ is contractible for every $x, y \in A$. These distinctions are the first instances of truncation. We will explain more about these ideas in Section 2.

2. Preliminaries

Category theory. In Section 1 we've seen that the primitive objects of study in category theory are animae.

Definition 2.1 (Truncated animae)**.** Let A be an anima.

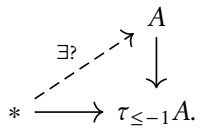
(a) A is (-2)-*truncated* if $A \simeq *$, i.e., if A is contractible;
(b) A is n-*truncated*,[2] for an integer $n \geq -1$, if $\mathrm{Eq}(x, y)$ is $(n-1)$-truncated for all $x, y \in A$.

Remark 2.2 (Truncation)*.*

(a) For $-2 \leq n \leq \infty$, let $\mathrm{An}_{\leq n} \subseteq \mathrm{An}$ be the full subcategory of n-truncated anima. Its inclusion has a left adjoint $\tau_{\leq n}$, called n-*truncation*.
(b) The (-1)-truncation functor $\tau_{\leq -1}$ is left adjoint to the inclusion of truth values. Explicitly, $\tau_{\leq -1}$ sends an anima to its set of connected components.

Remark 2.3 (Type theoretical interpretation)*.* In type theory, *propositions* are (-1)-truncated anima. To prove a proposition is to construct a term of that type. For an anima A, the truth value $\tau_{\leq -1}A$ expresses that A is nonempty. An *term* or an *element* of $\tau_{\leq -1}A$ is a proof of this proposition.

Constructing a term of A gives such a proof, but a proof of mere existence does not in general specify a term of A. Type theory thus makes explicit the distinction between proving existence and exhibiting a witness. More precisely, recovering a witness from a proof of existence is a lifting problem:



The lifts themselves form an anima.

Remark 2.4*.* Most of the animae are not be n-truncated for any $n \geq -1$, even when obtained by simple constructions. For example, define the n-*sphere* S^n inductively by

$$S^{-1} = \varnothing \quad \text{and} \quad S^n = \Sigma S^{n-1}$$

where $n \geq 0$ and $\Sigma X = * \sqcup_X *$ is the *suspension* of X, where the pushout is taken in An.[3]

In particular, S^0 consists of two points, and S^1 is obtained by adjoining two identifications between two points. For every $n \geq 2$, the sphere S^n is not m-truncated for any finite $m \geq -1$.

Exercise 2.5*.* The map $\eta \colon S^3 \to S^2$ called the *Hopf fibration*. How can one interpret the equivalence $\tau_{\leq 0}\,\mathrm{Map}(S^3, S^2) \simeq \mathbf{Z}\eta$, where η is the *Hopf element*, from the perspective of type theory? What does η represent in the corresponding type, and why does it exist?

Definition 2.6. A functor $f \colon \mathcal{C} \to \mathcal{D}$ is called

(a) *fully faithful* if the canonical map

$$\mathrm{Map}_{\mathcal{C}}(x, y) \longrightarrow \mathrm{Map}_{\mathcal{D}}(f(x), f(y))$$

is an equivalence for all $x, y \in \mathcal{C}$;
(b) a *localization* if, for every category $\mathcal{E}$, precomposition with f identifies $\mathrm{Fun}(\mathcal{D}, \mathcal{E})$ with the full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathcal{E})$ consisting of functors sending W to equivalences. We then write $\mathcal{D} \simeq \mathcal{C}[W^{-1}]$.

Proposition 2.7. *Let $f \colon \mathcal{D} \to \mathcal{C}$ be a functor with a left or right adjoint $g \colon \mathcal{C} \to \mathcal{D}$. The following are equivalent:*

(a) f is fully faithful;
(b) g is a localization.

[2] An n-truncated anima is also called an n-*groupoid*.
[3] Intuitively, ΣX is obtained from two points by adjoining identifications between them parametrized by X.

Proof. exercise. □

Definition 2.8 (Bousfield localization)**.** In Proposition 2.7, assume that f is fully faithful. We call $\mathcal{D}$ a *left Bousfield localization* of $\mathcal{C}$ if g is left adjoint to f, and a *right Bousfield localization* if g is right adjoint to f.

Convention 2.9. Consider the following adjunction diagram, i.e., f is the left adjoint of g:

$$\mathcal{C} \underset{g}{\overset{f}{\rightleftarrows}} \mathcal{D}.$$

We will always write the left adjoint on top of the right adjoint.

Example 2.10 (Bousfield localizations)*.*

(a) The inclusion $\mathrm{Ab} \hookrightarrow \mathrm{Grp}$ of abelian groups into groups has a left adjoint, called *abelianization*. Thus, Ab is a left Bousfield localization of Grp.
(b) The inclusion $\mathrm{Ab} \hookrightarrow \mathrm{CMon}$ has both a left and a right adjoint. Thus, Ab is both a left and a right Bousfield localization of CMon.
(c) The inclusion $\mathrm{Mod}_{\mathbf{Q}} \hookrightarrow \mathrm{Ab}$ has both a left and a right adjoint. Thus, $\mathrm{Mod}_{\mathbf{Q}}$ is both a left and a right Bousfield localization of Ab.

Exercise 2.11. Show that a fully faithful functor is a monomorphism in Cat and that a localization is an epimorphism. Do the converses hold?

3. Localization and purity

4. Classifying spaces

5. Vector bundles

6. Motivic spectra

7. Examples

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